CHE 210 Spring 2020 Quiz #4

Overall Balance

$A_4 = A_1 - \dot{\xi}_1 \Rightarrow \dot{\xi}_1 = A_1 - A_4 = (100-20)\frac{\text{mol}}{\text{min}} = 80 \text{ mol/min}$

$B_4 = B_1 - \dot{\xi}_1 - \dot{\xi}_2 = \dot{\xi}_2 = B_1 - B_4 - \dot{\xi}_1 = (150-10-80) = 60 \text{ mol/min}$

$P_4 = \dot{\xi}_1 = 80 \text{ mol/min}$

$Q_4 = \dot{\xi}_2 = 60 \text{ mol/min}$

Splitter Balance

$A_3 = A_4 + A_5 \Rightarrow A_5 = A_3 - A_4 = 145 - 20 = 125 \text{ mol/min}$

$B_3 = B_4 + B_5 \Rightarrow B_5 = B_3 - B_4 = 72.5 - 10 = 62.5 \text{ mol/min}$

$P_3 = P_4 + P_5 \Rightarrow P_3 = (80+500) = 580 \text{ mol/min}$

$Q_3 = Q_4 + Q_5 \Rightarrow Q_5 = Q_3 - Q_4 = 435 - 60 = 375 \text{ mol/min}$

split fraction $y = \frac{P_4}{P_3} = \frac{80}{580} = \frac{A_4}{A_3} = \frac{B_4}{B_3} = \frac{Q_4}{Q_3}$

$\therefore A_3 = A_4\left(\frac{P_3}{P_4}\right) = 20\left(\frac{580}{80}\right) = 145 \text{ mol/min}$

$B_3 = B_4(P_3/P_4) = 10\left(\frac{580}{80}\right) = 72.5 \text{ mol/min}$

$Q_3 = Q_4(P_3/P_4) = 60\left(\frac{580}{80}\right) = 435 \text{ mol/min}$

Mixer Balance

$A_2 = A_1 + A_5 = 100 + 125 = 225 \text{ mol/min}$

$B_2 = B_1 + B_5 = 150 + 62.5 = 212.5 \text{ mol/min}$

$P_2 = P_5 = 580 \text{ mol/min}$

$Q_2 = Q_5 = 375 \text{ mol/min}$

Fractional Conversion

$f_A^{ov} = \frac{A_1 - A_4}{A_1} = \frac{100-20}{100} = 0.800$

$f_A^{sp} = \frac{A_2 - A_3}{A_2} = \frac{225-145}{225} = 0.355$

Frac Selectivity

$\frac{\text{moles of B to desired Prod}}{\text{mole of B reacted}} = \frac{(A_2 - A_3)\frac{\nu_B}{\nu_A}}{(B_2 - B_3)} = \frac{(225-145)\left(\frac{1}{1}\right)}{(212.5-72.5)} = \frac{80}{140}$

$= 0.571$